

Title: Azure Cloud Fundamentals and Data Pipeline
Implementation using ADF

Internship: Data Engineering Internship

Week: Week 4 Assignment

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Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

Task 1: Azure Portal & Resource Group

1. Logged into Microsoft Azure portal.
2. Opened **Resource Groups**.
3. Clicked **Create**.
4. Entered resource group name as **rg-week4-jatin**.
5. Selected region as **Central India**.
6. Created the resource group successfully.

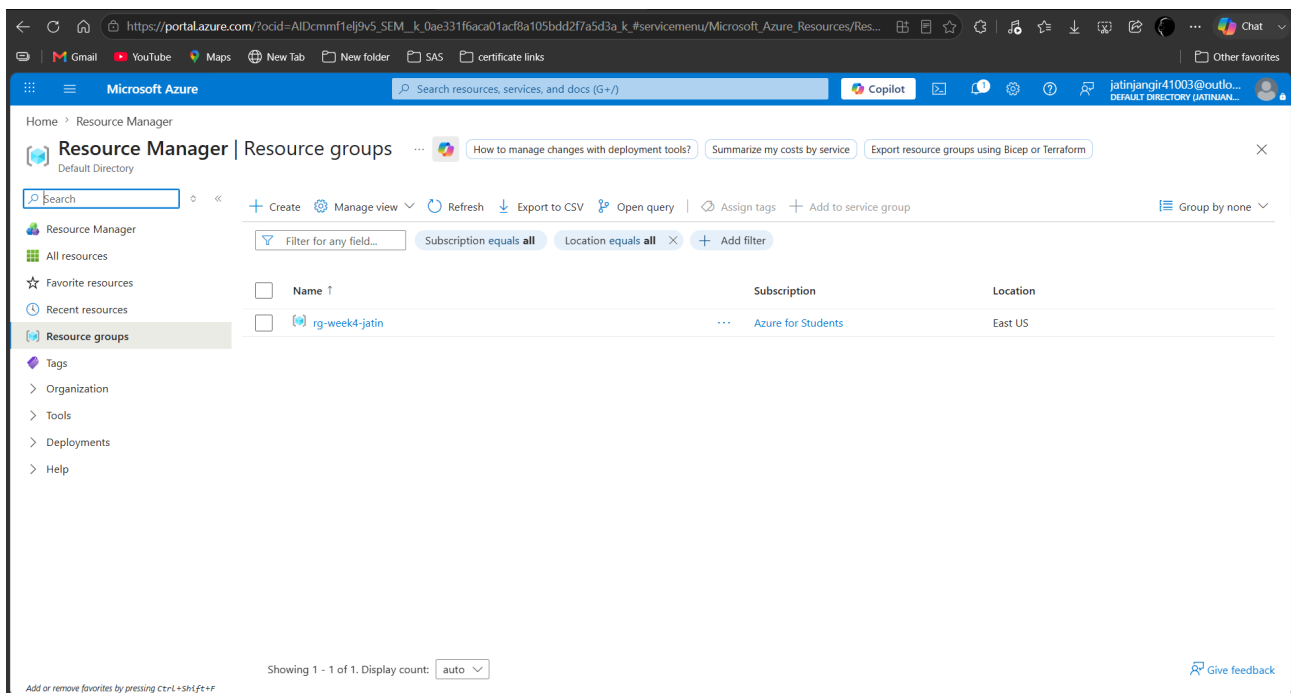


Figure 1: Resource Group Creation

Task 2: Storage Setup

1. Opened **Storage Accounts**.
2. Clicked **Create**.
3. Entered storage account name as **jatinstorage2026**.
4. Selected:
 - Performance = Standard
 - Redundancy = LRS
5. Created storage account.
6. Open **Containers**.
7. Created blob container **superstore-data**.
8. Uploaded **superstore_data.csv**.

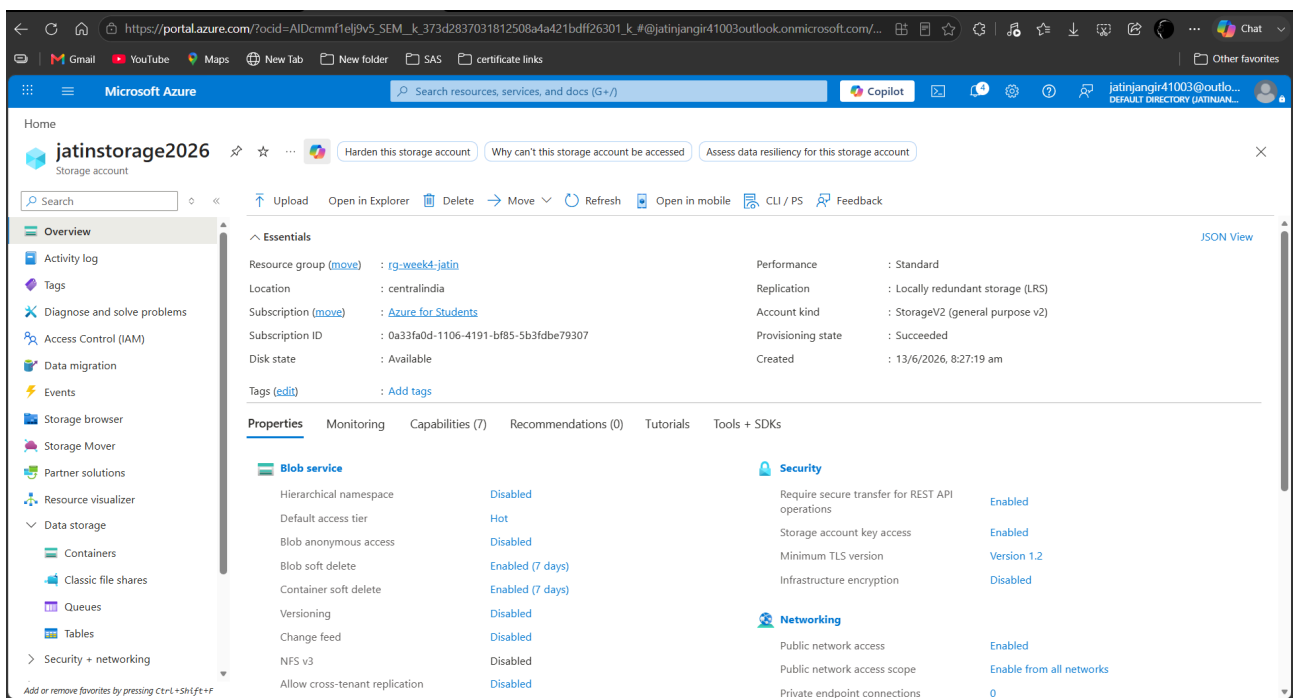


Figure 2: Storage Account Creation

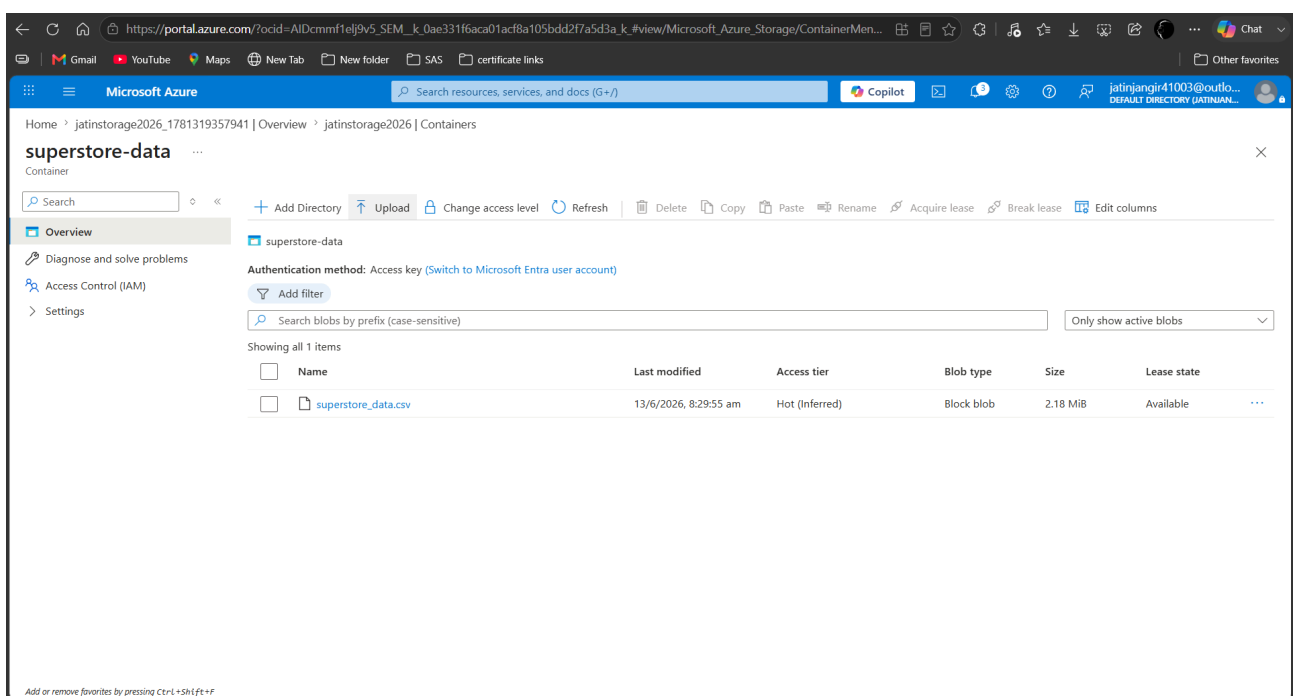


Figure 3: Container with Uploaded CSV

Task 3: Azure Data Factory Setup

1. Opened **Data Factories** in Azure.
2. Created a data factory named **adf-jatin-week4**.
3. Opened ADF Studio.
4. Explored:
 - Author
 - Monitor

○ Manage

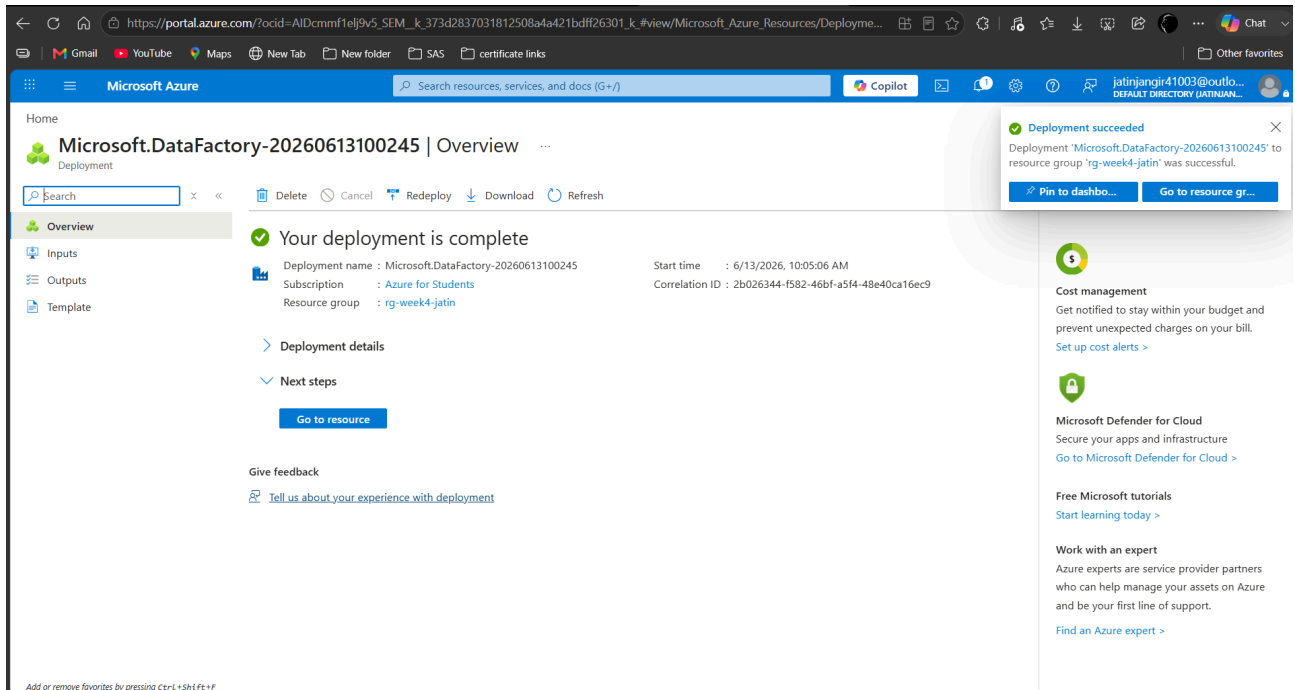


Figure 4: ADF Created

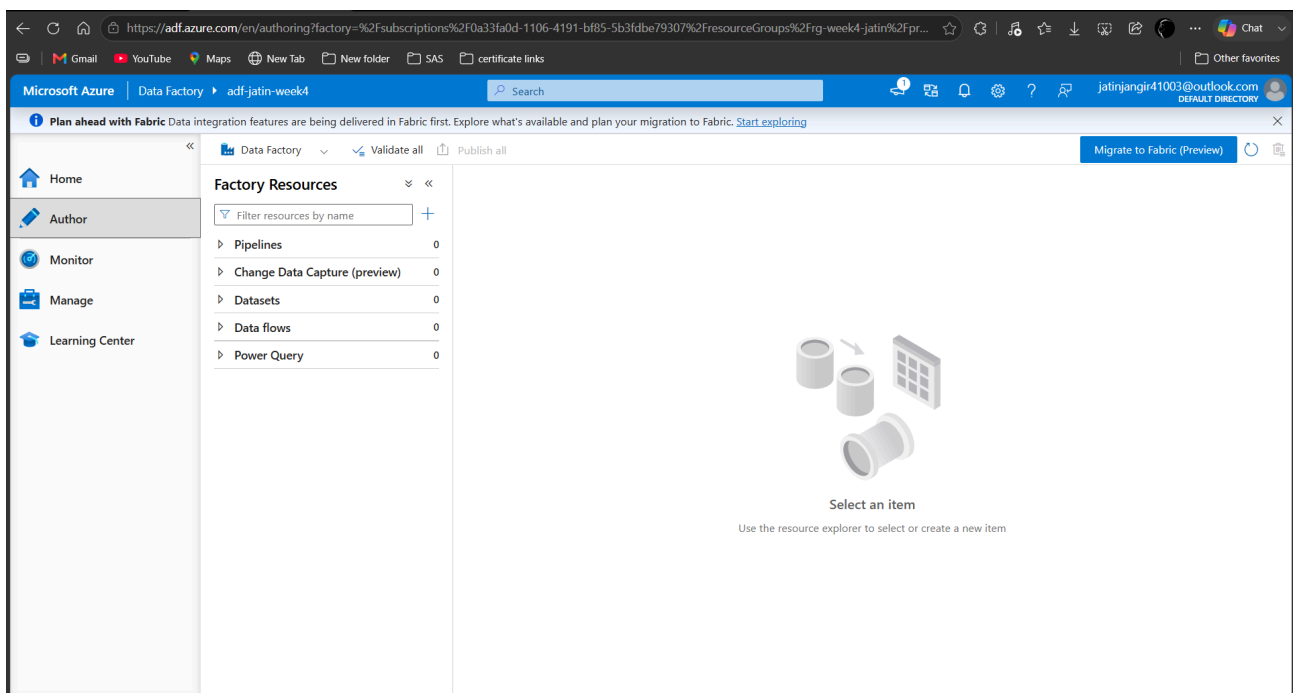


Figure 5: ADF Explore

Linked Service & Datasets

1. Opened **Manage** → **Linked Services**.
2. Created linked service **AzureBlobStorage1**.

3. Connected blob storage with ADF.
4. Created source dataset:
 - Name = **source_dataset**
5. Created destination dataset:
 - Name = **destination_dataset**

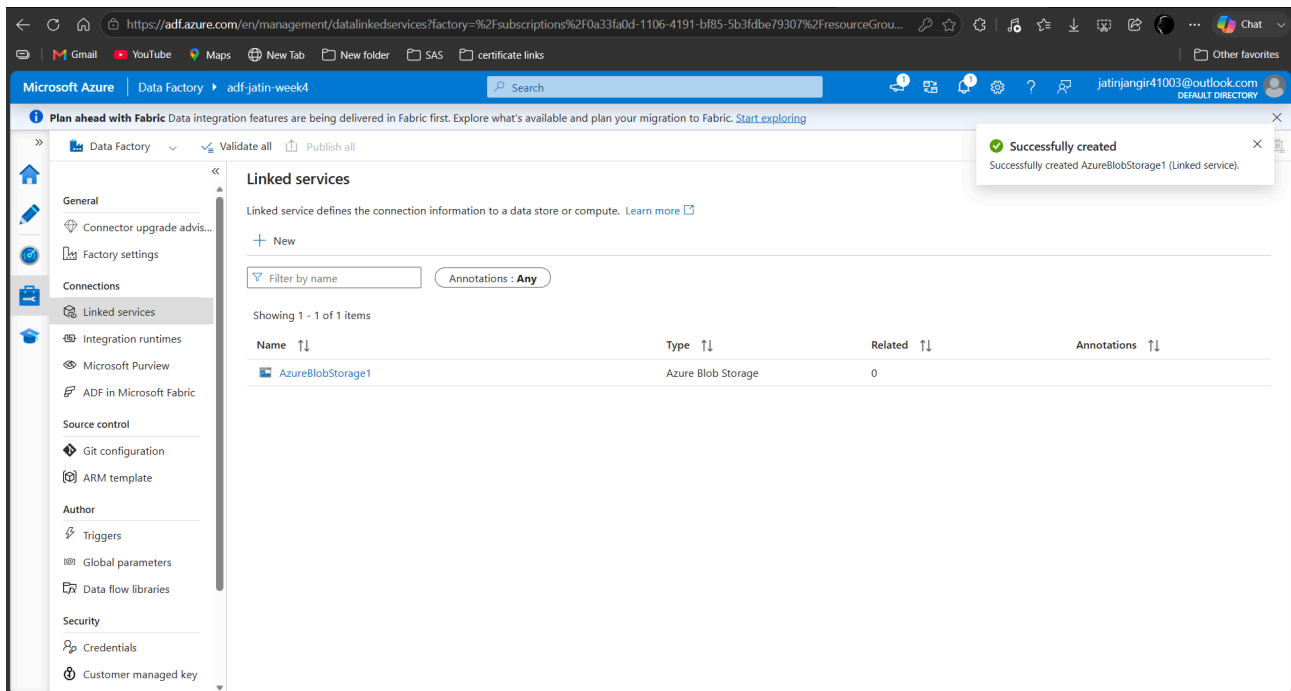


Figure 6: Linked Service

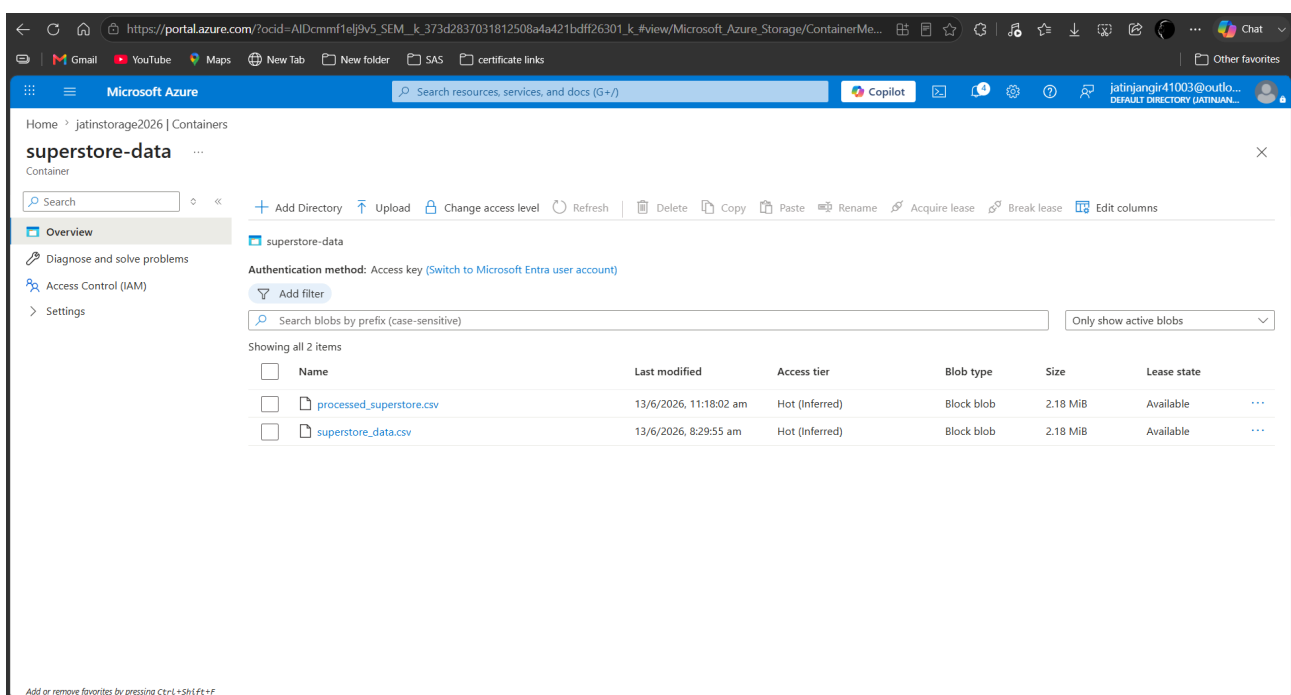


Figure 7: Source Dataset and Destination Dataset

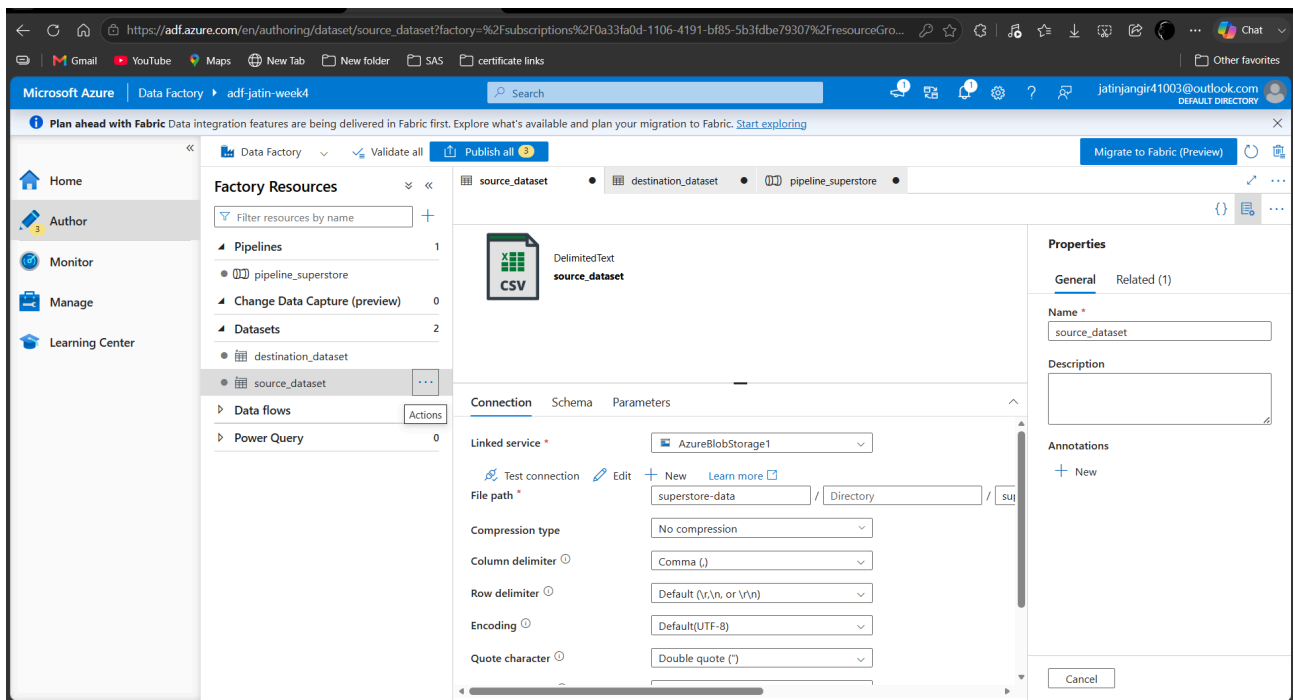


Figure 8: Source Dataset

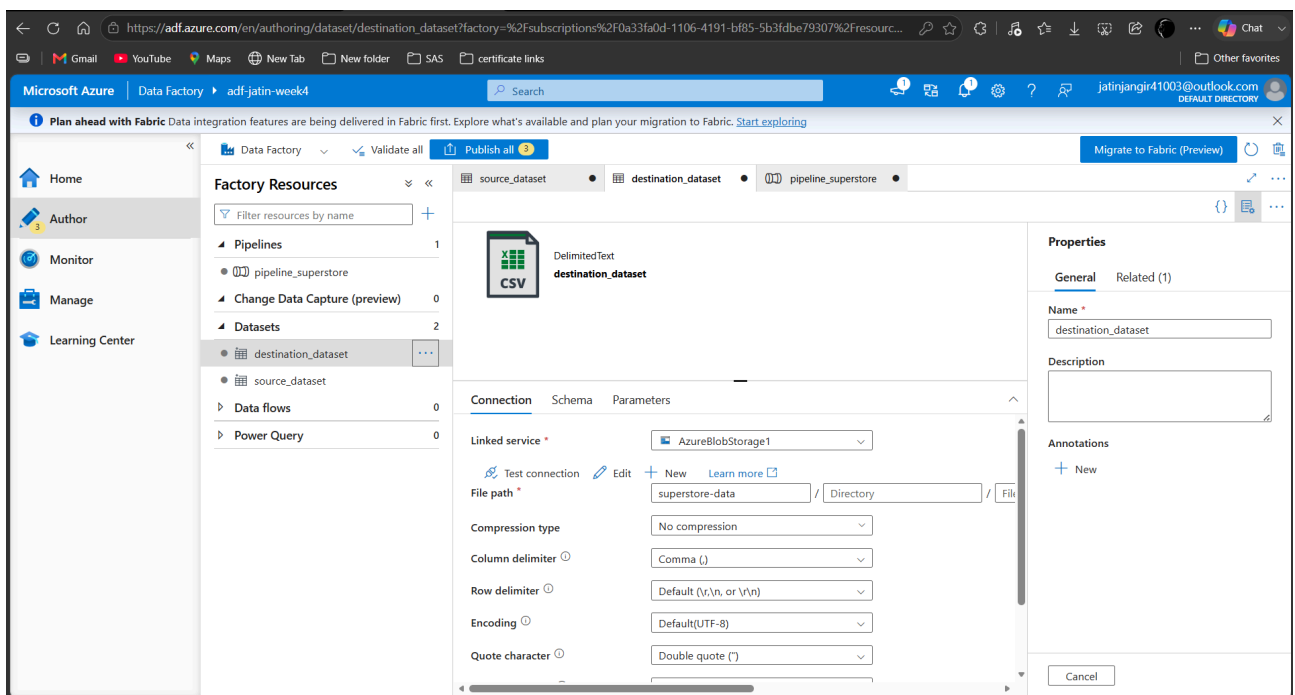


Figure 9: Destination Dataset

Get Metadata Activity

1. Created pipeline **pipeline_superstore**.
2. Added **Get Metadata** activity.

3. Connected dataset **source_dataset**.
4. Selected metadata fields:
 - Exists
 - Size
 - Last Modified

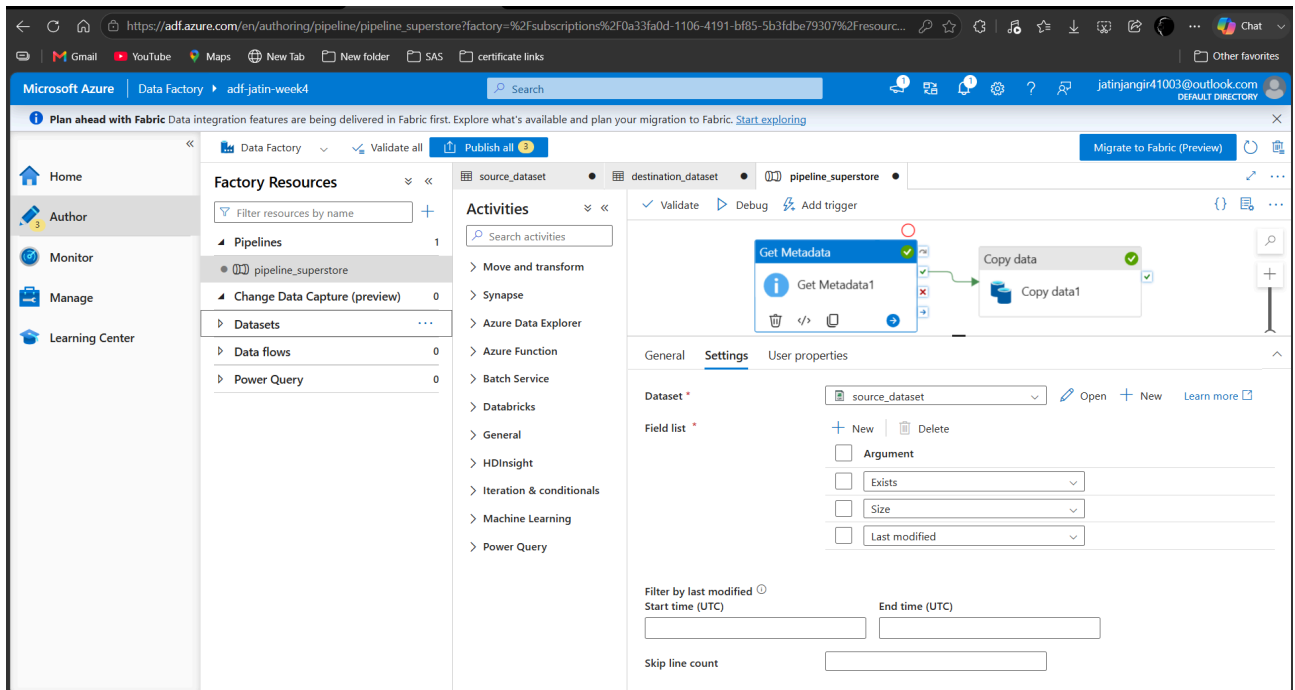


Figure 10: Get Metadata Activity

Pipeline Development

1. Added **Copy Data** activity.
2. Connected:
 - Get Metadata → Copy Data
3. Configured source dataset.
4. Configured sink dataset.
5. Set copy behavior.

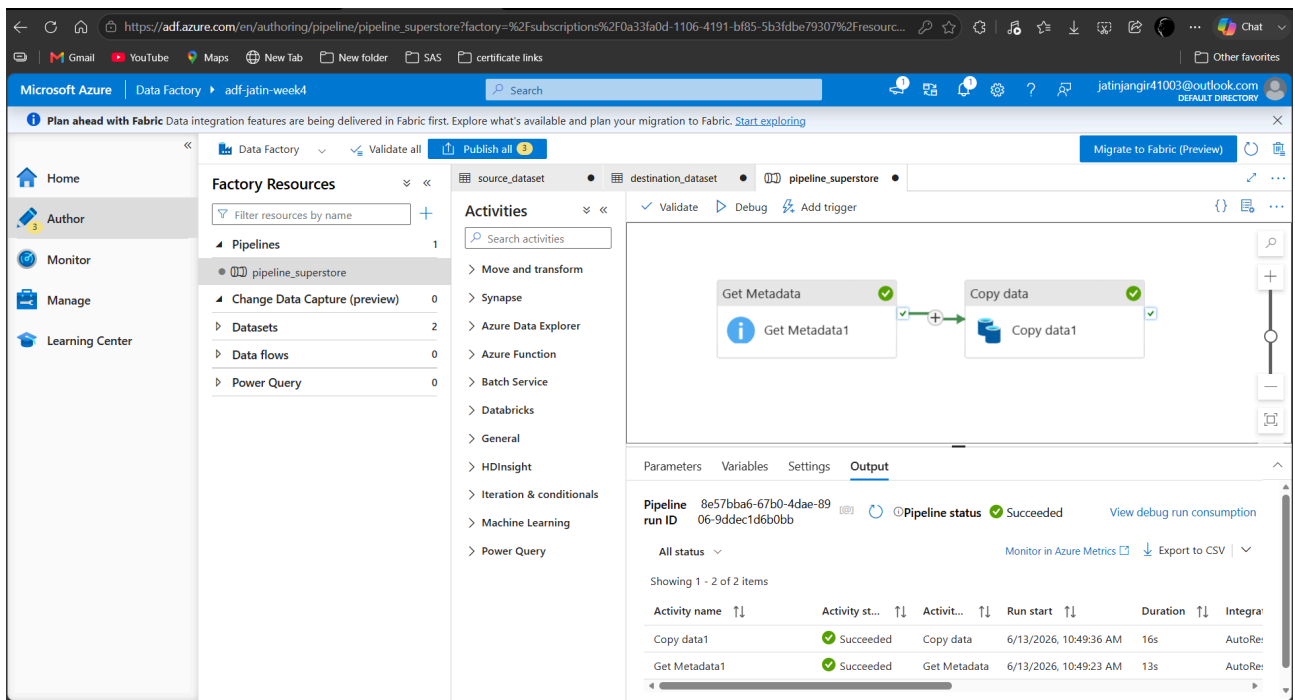


Figure 11: Pipeline Design

Pipeline Execution

1. Validated pipeline.
2. Clicked **Publish All**.
3. Ran pipeline using **Debug / Trigger Now**.
4. Monitored execution in **Monitor** tab.
5. Pipeline status showed **Succeeded**.

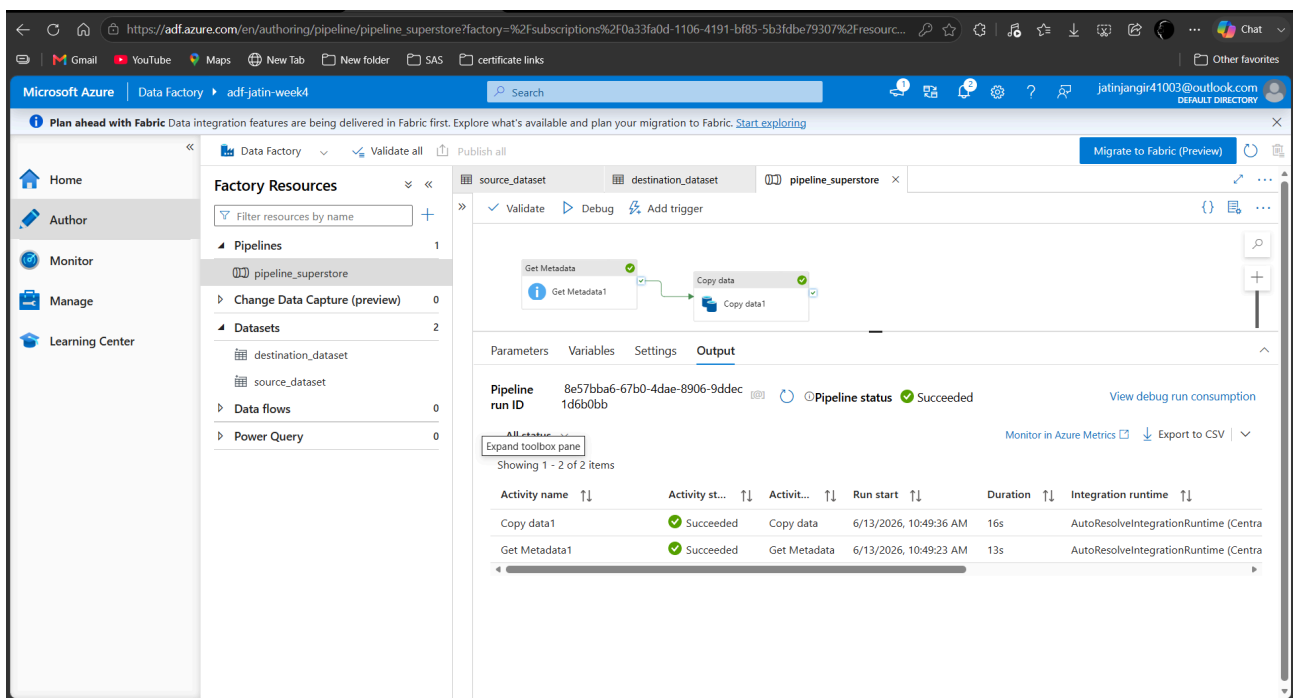


Figure 12: Pipeline Succeeded

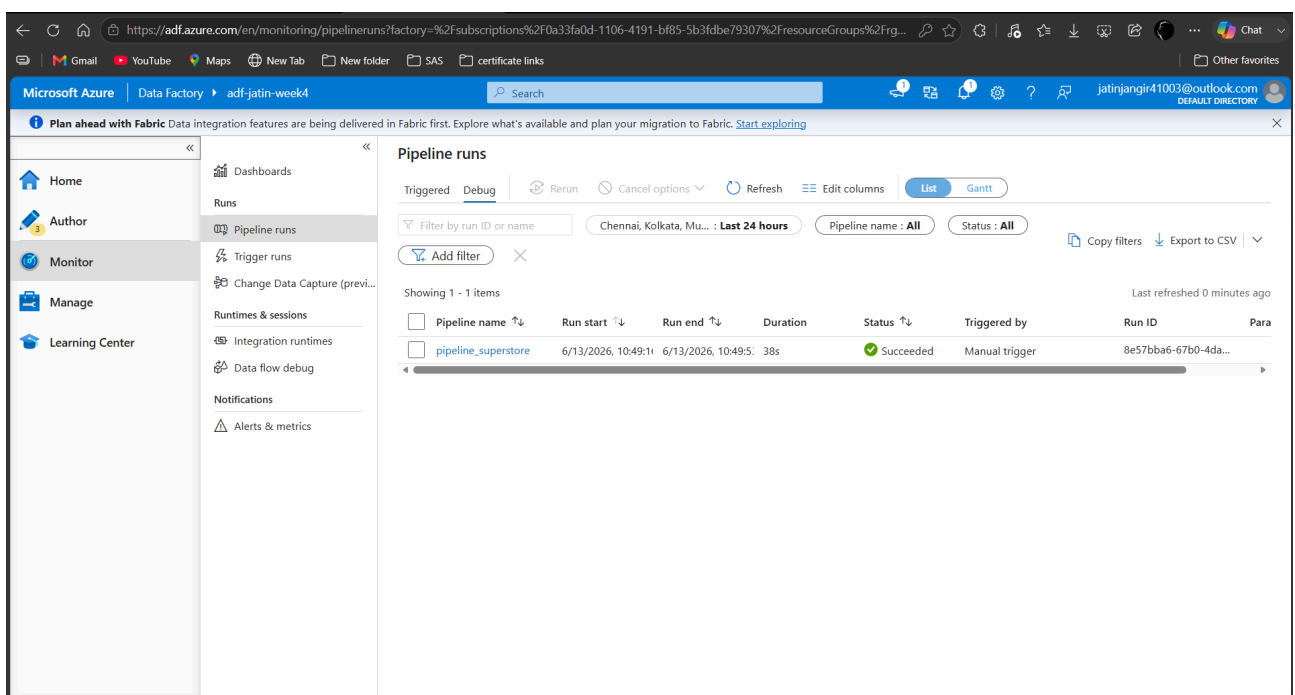


Figure 14: Monitor Tab

IAM Roles

1. Open Storage Account.
2. Open **Access Control (IAM)**.
3. Assigned:
 - Contributor role
 - Reader role
4. Verified successful assignment.

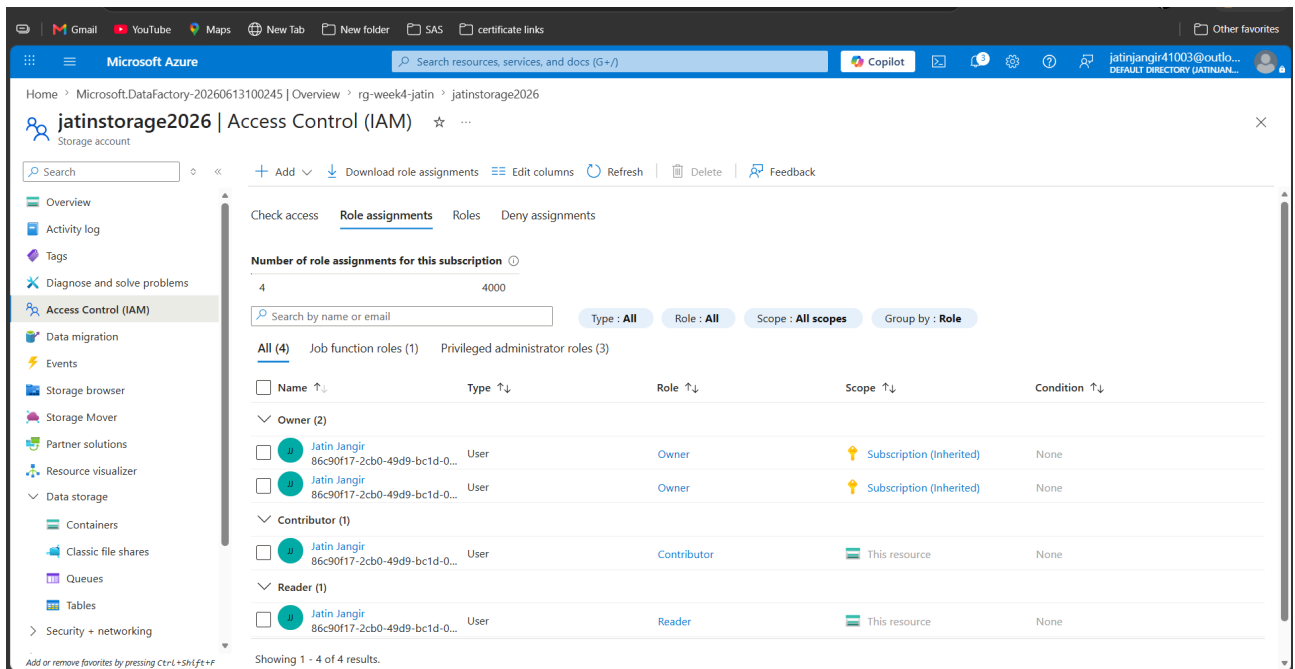


Figure 15: Access Control (IAM) Contributor role and Reader role